

FOCUS ON...

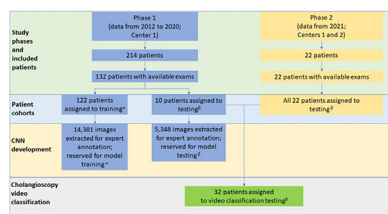
ORIGINAL ARTICLE

Identification of visible lesions during surveillance endoscopy for Barrett's esophagus: a video-based survey study

Claire A. Beveridge, MD, Chetan Mittal, MD, V. Raman Muthusamy, MD, MAS, Amit Rastogi, MD, Vladimir Kushnir, MD, Mariah Wood, MPH, Sachin Wani, MD, Srinadh Komanduri, MD, MS

Visible lesion (VL) detection is essential in patients with Barrett's esophagus (BE). The authors sought to assess the rate of VL detection by academic and community endoscopists using high-definition white-light endoscopy (HD-WLE) and narrow-band imaging (NBI) during surveillance endoscopy. Fifty endoscopists were invited to participate in a prospective video survey study. Participants viewed 25 standardized clips of patients referred for endoscopic therapy. Participants noted identification of anatomic landmarks and VLs using HD-WLE and NBI and reported practice-level data. The criterion standard of VL identification was established by consensus of 5 BE experts. The primary outcome was the rate of VL identification using HD-WLE and NBI. Forty-four of 50 participants completed the study (22 academic and 22 community). Compared with the criterion standard, participants did not identify 28% (HD-WLE) and 31% (NBI) of VLs. Community endoscopists had more experience (> 5 years in practice: community 85% vs academic 54.5%, $P = .041$; > 5 surveillance endoscopies a month: community 85% vs academic 31.8%, $P = .046$). Across all participants, VL detection using NBI improved significantly with a minimum of 5 surveillance endoscopies per month (area under the curve = .72; 95% confidence interval, .56-.85; $P = .006$). Despite improved endoscope resolution and availability of virtual chromoendoscopy, the overall rate of VL detection remains low. Identification of VLs using NBI may be volume dependent. Further education and training efforts focused on VL detection during BE surveillance endoscopy are needed.

Read this article on pages 241-7 in this issue.



ORIGINAL ARTICLE

Identification of patients with malignant biliary strictures using a cholangioscopy-based deep learning artificial intelligence (with video)

Neil B. Marya, MD, Patrick D. Powers, Bret T. Petersen, MD, Ryan Law, DO, Andrew Storm, MD, Rami R. Abusaleh, MD, Prashanth Rau, MD, Courtney Stead, MD, Michael J. Levy, MD, John Martin, MD, Eric J. Vargas, MD, Barham K. Abu Dayyeh, MD, MPH, Vinay Chandrasekhara, MD

Accurate diagnosis of malignant biliary strictures as benign or malignant remains challenging. It has been suggested that direct visualization and interpretation of cholangioscopy images have greater accuracy for stricture classification than current sampling techniques (eg, brush cytology and forceps biopsy) via ERCP. The authors aimed to develop a convolutional neural network (CNN) model capable of accurate stricture classification and real-time evaluation based solely on cholangioscopy image analysis. Consecutive patients with cholangioscopy examinations from 2012 to 2021 were reviewed. A CNN was developed and tested using cholangioscopy images with direct expert annotations. The CNN was then applied to a multicenter, reserved test set of cholangioscopy videos. CNN performance was then directly compared with that of ERCP sampling techniques. Occlusion block heatmap analyses were used to evaluate and rank cholangioscopy features associated with malignant biliary strictures. A total of 154 patients with available cholangioscopy examinations were included in the study. The final image database comprised 2,388,439 still images. The CNN demonstrated good performance when tasked with mimicking expert annotations of high-quality malignant images (area under the receiver operating characteristic curve = 0.941). Overall accuracy of CNN-based video analysis (0.906) was significantly greater than that of brush cytology (0.625, $P = .04$) or forceps biopsy (0.609, $P = .03$). Occlusion block heatmap analysis demonstrated that the most frequent image feature for a malignant biliary stricture was the presence of frond-like mucosa/papillary projections. This study demonstrates that a CNN developed using cholangioscopy data alone has greater accuracy for biliary stricture classification than traditional ERCP-based sampling techniques.

Read this article on pages 268-78 in this issue.